

9.1 Current & Potential Difference

Question Paper

Course	CIEA Level Physics
Section	9. Electricity
Topic	9.1 Current & Potential Difference
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What describes the electric potential difference between two points in a wire that carries a current?

- A. the force required to move a unit positive charge between the points
- B. the ratio of the energy dissipated between the points to the current
- C. the ratio of the power dissipated between the points to the current
- D. the ratio of the power dissipated between the points to the charge moved

[1 mark]

Question 2

When there is **no** current in a wire, which statement about the conduction electrons in that wire is correct?

- A. electrons in the wire are moving totally randomly within the wire
- B. equal numbers of electrons move at the same speed, but in opposite directions, along the wire
- C. no current is flowing therefore the electrons in the wire are stationary
- D. no current is flowing therefore the electrons in the wire are vibrating around a fixed point

[1 mark]

Question 3

A filament lamp has a resistance of $180\ \Omega$ when the current in it is $500\ \text{mA}$.

What is the power transformed in the lamp?

- A. $45\ \text{W}$
- B. $50\ \text{W}$
- C. $90\ \text{W}$
- D. $1400\ \text{W}$

[1 mark]

Question 4

When will 1 C of charge pass a point in an electrical circuit?

- A. when 1 A moves through a potential difference of 1 V
- B. when a power of 1 W is used for 1 s
- C. when the current is 5 mA for 200 s
- D. when the current is 10 A for 10 s

[1 mark]

Question 5

There is a current of 10 mA in a conductor for half an hour.

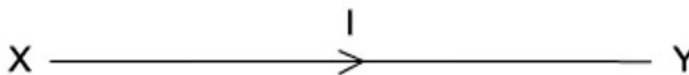
How much charge passes a point in the conductor in this time?

- A. 0.3 C
- B. 5 C
- C. 18 C
- D. 300 C

[1 mark]

Question 6

The diagram shows the symbol for a wire carrying a current I



What does this current represent?

- A. the amount of charge flowing past a point in XY per second
- B. the number of electrons flowing past a point in XY per second
- C. the number of positive ions flowing past a point in XY per second
- D. the number of protons flowing past a point in XY per second

[1 mark]

Question 7

A 12 V battery is charged for 20 minutes by connecting it to a source of electromotive force (e.m.f.). The battery is supplied with 7.2×10^4 J of energy in this time.

How much charge flows into the battery?

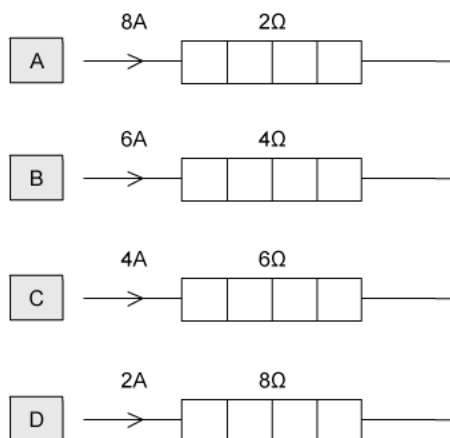
- A. 5.0 C
- B. 60 C
- C. 100 C
- D. 6000 C

[1 mark]

Question 8

The diagram shows four heaters and the current in each.

Which heater has the greatest power dissipation ?



[1 mark]

Question 9

Which equation that links some of the following terms is correct?

potential difference (p.d.) V

current I

resistance R

charge Q

energy E

power P

time t

A. $P = \frac{Q^2 R}{t}$

B. $ER^2 = V^2 t$

C. $\frac{VI}{P} = t$

D. $PQ = EI$

[1 mark]

Question 10

A copper wire of cross-sectional area 2.0 mm^2 carries a current of 10 A.

How many electrons pass through a given cross-section of the wire in one second?

A. 1.0×10^1

B. 5.0×10^6

C. 6.3×10^{19}

D. 3.1×10^{25}

[1 mark]

Model Answers: Easy

1

The correct answer is **C** because:

- Power is equal to $P = VI$
 - So, potential difference is equal to $V = \frac{P}{I}$
- This equation tells us that potential difference between two points in a wire is equal to the power dissipated between the points divided by the current flowing in the wire
 - This is also known as the ratio of these two quantities

A is incorrect because this describes the work done in moving a charge, which is not the same as the power dissipated

B is incorrect because power and energy are different quantities

D is incorrect because current and charge are different quantities

2

The correct answer is **A** because:

- Even when there is no potential difference applied through a metal wire, the free electrons are still moving around rapidly
- Without the potential difference, the speed of the electrons is about 10^6 m s^{-1}
- However, since they are moving at random, there is no net flow of electrons in any particular direction and so there is no current
- When a voltage is applied, the electrons gain an additional velocity, so that there is a net flow along the wire

B is incorrect because when there is no current flow in a wire, free electrons in a metal move around with a range of speeds and travel in random directions, much like molecules of gas in the air

C is incorrect because the current is the flow of electrons, but when there is no potential difference, there is no net flow of electrons, this doesn't mean the electrons are stationary, it just means they are not all moving in a specific direction – this distinction is very important to learn

D is incorrect because the outer electrons in a metal are not confined to a fixed point, they are delocalised and are able to drift through the metal freely

Exam tip: a common misconception is that electrons orbit the individual metal atoms, but this is not true. In reality, the outermost electrons are orbiting among all the metal atoms all the time, in a sort of “sea of electrons”. Another misconception is that electrons *only* jump between atoms when a voltage is applied along the wires, this is also not true. As we discussed in the question, the electrons in the “sea of electrons” are moving randomly inside the metal all the time. Applying the voltage causes the electrons to start drifting towards the positive terminal, which is what leads to a flow of electrons, better known as the current.

3

The correct answer is **A** because:

- Power is equal to $P = VI$
- Potential difference, current and resistance are related by the equation $V = IR$
- We are given current and resistance, so we need to eliminate V
- Combining the two equations gives
 - $P = (IR)I = I^2R$
- So, $P = (500 \times 10^{-3})^2 \times 180 = 45 \text{ W}$
- Therefore, the power transformed in the lamp is 45 W

4

The correct answer is **C** because:

- Charge is defined by $Q = It$
- In option C, $I = 5 \times 10^{-3} \text{ A}$ and $t = 200 \text{ s}$
 - So, $Q = (5 \times 10^{-3}) \times 200 = 1 \text{ C}$

A is incorrect because in order to calculate charge, we need current and time, current and potential difference alone aren't enough

B is incorrect because power and time can be used to calculate energy, not charge

D is incorrect because this gives $Q = 10 \times 10 = 100 \text{ C}$

5

The correct answer is **C** because:

- Charge is defined by $Q = It$
 - Current I is equal to $10 \times 10^{-3} \text{ A}$
 - The time t is equal to 30 minutes = $60 \times 30 = 1800 \text{ s}$
- Charge is therefore equal to $Q = (10 \times 10^{-3}) \times 1800 = 18 \text{ C}$

6

The correct answer is **A** because:

- Electric current is the rate of flow of charge
- It is defined as the amount of charge that passes through a point on a wire per unit time
- The only statement that matches this definition is option A

B, C & D are incorrect because electrons are the most common carriers of charge in electricity, but current is not specifically the flow of electrons, it can be the flow of any particle which carries a charge, whether it's electrons, protons or ions

7

The correct answer is **D** because:

- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
 - In equation form this can be expressed as $V = \frac{E}{Q}$
 - Rearranging for charge Q gives $Q = \frac{E}{V}$
- Therefore, the charge that flows is $Q = \frac{7.2 \times 10^4}{12} = 6000 \text{ C}$

8

The correct answer is **B** because:

- Power is equal to $P = VI$
- Potential difference, current and resistance are related by the equation $V = IR$
- We are given current and resistance, so we need to eliminate V
- Combining the two equations gives
 - $P = (IR)I = I^2 R$
- Let's consider option B
 - Current $I = 6 \text{ A}$, Resistance $R = 4 \Omega$
 - $P = 6^2 \times 4 = 144 \text{ W}$
- This is the highest power dissipation out of all the options (see other calculations below)

A is incorrect because $P = I^2 R = 8^2 \times 2 = 128 \text{ W}$

C is incorrect because $P = I^2 R = 4^2 \times 6 = 96 \text{ W}$

D is incorrect because $P = I^2 R = 2^2 \times 8 = 32 \text{ W}$

9

The correct answer is **D** because:

- Power is equal to $P = \frac{E}{t}$
- Charge is equal to $Q = It$
- The units on the LHS
 - $PQ = [W][C] = [Js^{-1}][As] = JA$
- The units on the RHS
 - $EI = [J][A] = JA$
- The units on the LHS match the units on the RHS so option D is correct

$$\text{LHS: } P = [W] = Js^{-1}$$

A is incorrect because

$$\text{RHS: } \frac{Q^2 R}{t} = \frac{(It)^2}{t} \frac{V}{I} = VIt = E = J$$

LHS $\neq$ RHS

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B is incorrect because

$$\text{LHS: } ER^2 = VIt \left(\frac{V}{I} \right)^2 = \frac{V^3 t}{I}$$

LHS $\neq$ RHS

C is incorrect because

$$\text{LHS: } \frac{VI}{P} = \frac{P}{P} = 1$$

LHS $\neq$ RHS

$$\text{RHS: } t$$

10

The correct answer is **C** because:

- Current is defined as the amount of electric charge per unit of time
 - It is equal to $I = \frac{Q}{t}$
- First, we must find the charge Q over the period of 1 second:
 - $Q = It = 10 \times 1 = 10 \text{ C}$
- The number of electrons is calculated by dividing the total charge by the charge of one electron
- The charge of one electron e is $1.6 \times 10^{-19} \text{ C}$
 - Number of electrons $= \frac{Q}{e} = \frac{10}{1.6 \times 10^{-19}} = 6.3 \times 10^{19}$